Prediction is not selection: a pre-registered study of design-time mechanism choice in author name disambiguation

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Abstract

Can a designer measure a property of their own data before building anything and learn which classification mechanism to build? On seven public author name disambiguation datasets the answer is mostly no, for two separate reasons. The first is structural. A rule assigning on an exact surface match is a strict sub-case of a retriever assigning from the single nearest surface form, whenever the similarity function gives its maximum only to identical strings, because a test string present in the history is then its own nearest neighbour. In the comparison most practitioners would run, the rule therefore cannot win at any cutoff, and we measured zero rule wins on 199,713 held-out author mentions in two metric families before we understood why. A wider form of the argument says where a precision-first rule cannot win. The second reason is empirical. On a redesigned pair in which neither mechanism contains the other, an exact-evidence rule against a k-nearest vote, neither pre-registered signal is actionable at the margin we fixed in advance once the seven datasets are pooled: the mechanisms sit half an accuracy point apart, 0.8244 against 0.8294, no band on either axis clears the margin, and both selection hypotheses return Inconclusive. The pooled result is mostly inspire's, which holds four fifths of the blocks, and it is not uniform. Two of the five datasets large enough to judge are decided at that margin in opposite directions: kisti for the rule by 0.1131, through better matching, and zbmath for the vote by 0.0631, through the rule's abstentions. One unregistered signal, block size, does carry a decided pooled verdict, and it is composition and abstention rather than matching quality. The consistency statistic, by contrast, predicts each mechanism's own accuracy at Spearman 0.51 and 0.83. Prediction is not selection.

Keywords: author name disambiguation; pre-registration; mechanism selection; equivalence testing; entity resolution; negative results

1 Introduction

1.1 The design-time question

Someone building a record-linking or disambiguation system starts with a history and no labels for the future. They can read everything already recorded, but they cannot evaluate anything on the decisions that have not been made yet, and the question they face is which mechanism to build.

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Predicting how hard a problem will be from properties of its data is well studied, and so is choosing an algorithm per instance. Algorithm selection [Rice, 1976, Smith-Miles, 2009, Barbudo et al., 2023] asks which member of a portfolio performs best on a given instance, and answers it by fitting a selector on measured per-instance performance. That measurement needs labels. The question here is narrower and earlier: can a designer measure a property of their own history, before building anything and before any labelled evaluation exists, and learn from it which of two mechanisms to build? A signal can answer the first question perfectly and the second not at all. Knowing a block is hard tells you both mechanisms will do badly there; it does not tell you which will do less badly, and that is the question a designer has to answer.

1.2 Why author name disambiguation

The two halves of our answer do not carry the same scope, and the distinction is load bearing. The structural half is an argument about mechanisms that assign from a record's own history. It uses no property of names, so it is a result about entity resolution, and we state it at that width: the same assumption sits under probabilistic record linkage [Fellegi and Sunter, 1969], under the reference treatments of the modern field [Christen, 2012], and under the blocking methods that scale it [Papadakis et al., 2020]. The empirical half is a measurement, and a measurement is only of what was measured, so every number below about how two mechanisms differ is the author name disambiguation instance of the question.

That empirical half needs a substrate where two block-level properties vary natively, separately, and visibly at design time. Author name disambiguation has exactly that: how many distinct people share a name and how many ways one person's name is written are different properties of the same block, and both are readable from the history alone. A clean matching benchmark can lack the first. We began on the WDC product-matching corpus [Primpeli et al., 2019] and stopped there: with product cluster as the key every cluster carries one category, so the consistency axis is a point mass, and with normalized title as the key 90.7% of recurring keys still sit at exactly 1.0. That is one benchmark and we measured no other. We use S2AND [Subramanian et al., 2021], which unifies eight prior disambiguation datasets into one format, ships a reference implementation, measures both properties itself as homonymity and synonymity, runs offline, and gives cross-dataset generalization at no extra cost. Seven of its eight datasets are usable here.

1.3 What we found, and what we claim

The answer is no, for two separate reasons, and a third result qualifies how far the no reaches.

The comparison most people would run can only go one way. A rule that assigns on an exact surface match is a strict sub-case of a retriever that assigns from the single nearest surface form, because a test form present in the history is its own nearest neighbour under any similarity that reserves its maximum for identical strings. The retriever takes it, resolves the same author, and is right everywhere the rule is right, whatever the cutoff, provided the two mechanisms share a tie-break. We measured zero rule wins on 199,713 held-out signatures in two metric families before we understood why. The wider form covers any pair of mechanisms that assigns an author from the block's own history: a precision-first rule cannot win where the history names one author. Where the history is written one way, the same holds for mechanisms that read surface forms only.

Where a contest is possible, there is almost nothing to select once the datasets are pooled. We built a pair in which neither mechanism contains the other, an exact-evidence rule against a k-nearest vote, pre-registered the analysis, and ran it. The pair is two-sided on held-out data, and the two mechanisms are then within half an accuracy point pooled, 0.8244 against

0.8294. The gap compresses along both design-time axes and changes sign on both, but no band clears the pre-registered five-point margin, so both selection hypotheses return Inconclusive. The one clause asking which signal drives whether the choice is decidable at all found surface stability, not consistency, decisively at $k = 5$ and weakly at $k = 2$. Section 7.8 gives the exceptions: two datasets decided at the margin, one each way.

One signal we did not register does carry a verdict, and it is not the finding it looks like. Block size is readable from the blocking pass alone, so it meets the same design-time test as the other two, and on blocks of three to eight signatures the vote is ahead by more than five accuracy points, which no band of either registered signal manages when the seven datasets are pooled. Two measured facts change what that means. At fixed consistency, block size never points to a different winner than consistency already points to, so it says how much the choice is worth rather than which choice to make. And the decided bands come from the rule declining to answer rather than from either mechanism resolving a name better: where the rule commits it is the more accurate arm in every band. Section 7.7 reports the verdict in full and then both qualifications.

Against all that, the consistency signal is not weak. It predicts each mechanism's own held-out accuracy at Spearman 0.51 and 0.83, positive with intervals excluding zero in every qualifying dataset. It works; it just does not answer the question a designer is asking.

The four results do not share a scope. The containment and its general form are proven from the mechanisms alone, using no property of author names and no property of our data, so they are contributions to entity resolution and are stated at that width. The other two are the author name disambiguation instance, measured on seven datasets with one pair of untrained lookups, and we do not offer them as facts about entity resolution. What we claim empirically is what the frozen analysis supports and no more: pooled over the seven datasets, no design-time signal we examined tells a designer which mechanism resolves a name better, and Section 7.8 reports the two datasets where the picture differs. We do not claim that no such signal exists for any mechanism pair anywhere, nor that the effects below the margin are zero, since they are small, they change sign, and the frozen rules declined to decide them. We claim nothing about a selector, because none was built, for the reason in Section 7.9, and nothing about learned disambiguation systems, since both mechanisms here are untrained lookups by design.

1.4 Contributions

The paper's contributions are the following.

1. **A structural result about mechanism comparison.** Assigning on an exact surface match is a strict sub-case of single-nearest retrieval, so the verdict that the rule wins is unreachable at any cutoff, in any metric family that gives its maximum only to identical strings, and the comparison cannot inform the choice it is run to settle. For any pair of mechanisms assigning from the block's own history, a precision-first rule cannot win where the history names one author, and for mechanisms that read surface forms only it cannot win where the history is written one way. The containment itself has precedent in the blocking literature; the unreachable verdict and its general form are what we add, and we measured the verdict as zero rule wins on 199,713 held-out signatures before we derived it (Section 4).

2. **A pre-registered study of design-time selection on real data.** Seven public author name disambiguation datasets, 197,479 held-out signatures, a frozen falsification table, a mechanism pair in which neither contains the other, and a winner construct that keeps demonstrated ties apart from undecided blocks. Neither pre-registered signal is actionable at the pre-registered margin, and both selection hypotheses return Inconclusive, pooled; two datasets

depart from the pooled picture, one in each direction, and are reported as such (Sections 6 and 7).

3. **A deflation of the one signal that does decide.** Block size, which we did not pre-register, is the only signal here whose own bands carry a decided verdict when all seven datasets are pooled. Two measurements on the same records show what they are: at fixed consistency block size never points to a different winner, and on the items where the rule commits it is the more accurate arm in every band (Section 7.7).

4. **Cautions for portfolio and benchmark design.** Count the items on which each mechanism is right and the other wrong before comparing them; report structurally empty regions as properties of the design; and do not calibrate a retriever for its own accuracy when the purpose is a comparison, because in both tunable families we probed that moves its parameter toward a limit in which the pair is nested (Sections 5.2 and 8.2).

2 Related work

2.1 Decision-focused evaluation

A line of work argues that a model's score under a general-purpose metric is not the quantity a decision-maker cares about. Hand [2009] shows that the area under the receiver operating characteristic (ROC) curve implicitly averages over a cost distribution that depends on the classifier being scored, and Hernández-Orallo et al. [2012] give a unified account in which a metric is a choice of threshold policy plus a cost distribution. Vickers and Elkin [2006] make the point operationally in clinical prediction, and Elmachtoub and Grigas [2022] push it into the training objective.

Information retrieval has a parallel line about comparisons that cannot inform rather than metrics that cannot. Armstrong et al. [2009] found a decade of reported ad-hoc retrieval gains that did not accumulate, because the baselines they were measured against were weak. This paper belongs to that lineage and its structural version is sharper: there the baseline could have been beaten and was chosen badly, here one of the two verdicts cannot occur at all, whatever the data.

Our gap sits one step earlier than any of these. They ask whether a measurement answers the decision question once the candidate models exist. We ask whether a measurement taken before either candidate is built answers a choice between them. The unit differs too: theirs is a model on a dataset, ours is a mechanism on a block, with tens of thousands of blocks in one corpus.

2.2 Author name disambiguation as the home domain

Author name disambiguation is a mature benchmark domain with its own vocabulary for exactly the two axes we need. S2AND [Subramanian et al., 2021] unifies eight datasets and defines homonymity and synonymity per block, which we use as external checks on our own two signals, and WhoIsWho [Chen et al., 2023] provides a larger benchmark with a competition around it. Surveys of the area [Ferreira et al., 2012] classify methods by the evidence they use, and string-based matching is a standing baseline rather than a straw man: Backes [2018] argues that name-matching and blocking choices dominate reported differences between disambiguation methods, which is a version of the concern this paper makes structural. How disambiguation output should be scored is itself an active question [Kim, 2019], and Section 6.3 departs from the usual answer for a reason it gives there. We use S2AND as substrate rather than as target: we propose no disambiguation method, and neither of our mechanisms is competitive with the systems this literature builds. They are small on purpose, for the reasons in Section 3.1.

2.3 Surface-form robustness, and the nearest prior result

Kim and Kim [2020] is the closest prior work on the surface axis. They ask how much author forenames contribute to disambiguation on four labelled datasets, and they do it by manipulation, keeping full forenames for a random share of name instances and initializing the rest, sweeping that share from 0 to 100% in steps of ten, and separately truncating every forename to its first n characters. A zero-parameter string-matching heuristic is compared against feature-based machine learning with agglomerative clustering whose threshold is chosen per block against the B-cubed F1 being scored [Amigó et al., 2009], and the reported unit is the dataset crossed with the manipulation setting.

Their headline is gap compression: as full forenames become more common, the margin by which machine learning beats string matching shrinks. It shrinks and does not cross, string matching never beating the best algorithmic arm on B-cubed F1 in any of their four datasets at any setting, and the two local crossings on component metrics do not survive into F1. Their practical advice, the closest they come to our question, is that a reader should characterize a dataset's name ambiguity before choosing a method, which stops short of a measurable signal or a procedure.

Four differences matter. Their surface axis is manipulated and ours is measured as found, so theirs is not available to a practitioner reading their own data; their unit is the dataset and ours is the block; their surface variable is an experimental arm and ours is a design-time signal; and their outcome is a gap magnitude, where ours is a winner construct with an equivalence margin and a distinct tie. Section 7.5 reproduces their compression shape at block level with measured variation, and the finding is theirs. They also treat only the initialization form of synonymy and exclude edit-distance and token-order variation, which our surface measure covers and our initialization-free control separates.

2.4 Adjacent and out of scope

Three neighbouring literatures are about a different decision. The reject option [Chow, 1970, Hendrickx et al., 2024] and learning to defer [Mozannar and Sontag, 2020] both choose, per item and at inference time, whether to answer at all; our abstention is a fixed property of one mechanism rather than a tunable policy, and our decision is taken once, at design time, about what to build. The third is learned entity matching, where a design space of deep models is fitted to labelled pairs [Mudgal et al., 2018]. Both mechanisms here are untrained lookups by design, so that line marks what this paper deliberately is not, rather than a baseline it competes against.

3 Problem setting and constructs

3.1 Blocks, and what design time means

A name block is a set of author mentions sharing a normalized name string. Each block is a small classification problem in its own right: a history of mentions already assigned to authors, and new mentions arriving that have to be assigned to one of them. A single corpus holds tens of thousands of these and they differ enormously, which is why the block, not the dataset, is our unit of analysis, and it is what lets the design-time question be asked many times over instead of once.

Design time here names a constraint on what can be computed rather than a stage in a project. Every signal we call design-time comes from the training half of a block and nothing else. A statistic needing the test items, or their labels, is not available when the decision is made, whatever its correlation with the outcome afterwards.

Both mechanisms are untrained lookups, by design. A practitioner deciding what to build from the history alone, before committing to a labelled training effort, is choosing among untrained lookups. Two of them also hold everything else fixed: neither arm has a training budget, a feature set or a model class the other lacks, so a winner difference cannot be a difference in what was spent on each arm. Section 9 returns to what the choice scopes out.

3.2 The two axes

Two properties of a block vary in author name disambiguation, vary separately, and are both visible at design time. **Consistency** is how many distinct people a name refers to; S2AND calls the inverse of it homonymity. We measure it as ADS, the author-dominance share: the fraction of the block's training signatures belonging to its single most frequent author, so a history naming one person scores 1.0. The acronym is carried over from the earlier study of Section 8.3. **Surface stability** is how consistently one person's name is written; S2AND calls the inverse of it synonymity. We measure the inverse directly, as SV, the surface variability: the expected normalized edit distance between two training signatures of the block. We measure it; we do not set it.

Their separability is load-bearing. If the two signals were the same quantity measured twice, any claim that one predicts accuracy and the other predicts the winner would be an artifact of how we had defined them. Measured across the 18,984 live blocks, a wider population than the 13,902 every reported result uses, they correlate at Spearman -0.2612, so roughly 7% of rank variance is shared, and each measure tracks its own S2AND counterpart strongly and the other weakly. The winner, separately, is defined only by labelled held-out accuracy, and neither signal appears in the definition of either mechanism or of the winner.

3.3 What it means to win a block

A mechanism wins a block when its held-out accuracy there is higher than the other's by more than a fixed margin, with the difference resampled, so that on all but the smallest test sets a win has to clear the margin with its whole interval; Section 7.4 reports where that protection fails. The margin is there because a designer choosing what to build does not act on a tenth of an accuracy point. We fixed it at five accuracy points in advance and report two smaller values beside it. The construct is equivalence testing [Lakens, 2017] applied one block at a time: a margin standing for the smallest difference worth acting on, an interval read against it, and practical equivalence concluded only when the interval falls inside. That literature carries the caveat we inherit, that the margin is a judgement and the reading moves with it.

Four labels follow, not two. The rule wins, the vote wins, the two are demonstrated equivalent within the margin, or the evidence does not settle it. The last two are different statements and we never merge them. A block whose interval sits entirely inside the margin has shown the mechanisms interchangeable there; a block whose interval straddles a margin boundary has shown nothing. Merging them is how a low-powered study comes to read as a positive finding of equivalence, and a paper whose result is largely made of ties has to be able to say which kind of tie it has.

4 A comparison only one side can win

4.1 The containment argument

Consider the comparison a practitioner reaches for first. Two mechanisms resolve a held-out name mention against the mentions already recorded under the same name block. A precision-first *rule*

looks for a training record whose surface form is identical to the test form, assigns the author carried by that form, and abstains when no such record exists. A coverage-first *retriever* scores every distinct training form against the test form with a string similarity function, takes the nearest one, and assigns its author. Rules against retrieval, exact against fuzzy. Which one should you build?

Suppose the test form occurs in the block's history. The similarity functions used here return their maximum, 1.0, for identical strings and for no other pair, so no other training form can tie the test form at the top. A retriever that takes the single nearest form therefore takes the test form itself, and if the two mechanisms share a tie-break rule over candidates carrying the same form, they resolve the same author. Whenever the rule fires, the retriever returns what the rule returns; when the rule abstains, the retriever answers anyway. The rule's set of correct predictions is a subset of the retriever's, so the rule cannot be right where the retriever is wrong, and under any winner definition based on accuracy the verdict "the rule wins" is not merely rare. It is unreachable.

The argument does not depend on the acceptance cutoff, since a cutoff cannot exceed 1.0 and so cannot exclude the identical form. It does depend on three things, and the first is easy to miss. The similarity has to reserve its maximum for identical strings. Normalized Levenshtein and Jaro-Winkler similarity do, scored on raw strings as they are here, and across every pair of distinct forms inside our blocks, 210,537 pairs, none scores 1.0 in either family. Token-sorted, token-set and partial-match ratios do not, and neither does any metric applied after case folding, so a retriever built on one of those can be led away from the identical form by its tie-break. The other two conditions are that the retriever takes the single nearest form and that both mechanisms draw their candidates from the same history.

4.2 What we measured

We did not derive the containment first and then confirm it. We pre-registered the whole analysis, froze the calibration constants, ran it, and read the degeneracy off an empty cell of the output. That ordering is why this section is a result and not a remark: the degeneracy is easy to follow once someone states it in a paragraph, and it was, in our hands, hard enough to see in advance that it survived a pre-registration detailed enough to fix a deterministic tie-break chain for both mechanisms.

On the whole evaluation split, 16,136 blocks and 199,713 held-out signatures, the number of signatures on which exact match is correct and fuzzy retrieval is incorrect is **zero**. Zero in the Levenshtein arm and zero in the Jaro-Winkler arm. That count runs over the whole split, because a signature on which the rule beats the retriever would count wherever it fell. The cutoff had been calibrated once on a disjoint split, landing at 0.50 in both metric families; no value on the pre-registered grid from 0.50 to 0.95 changes the count, because none of them can.

Two pre-registered diagnostics collapsed along with the winner, both measured on the primary analysis population of Section 6.1, 13,902 blocks and 197,479 test signatures, which is what every reported hypothesis number uses. The pooled accuracy gap there is -0.0196, exact match 0.8244 against fuzzy retrieval 0.8441, and it is a coverage effect and nothing else: every signature exact match gets right fuzzy retrieval also gets right, so the gap is the share of test signatures on which exact match abstains and fuzzy retrieval is correct. Exact coverage on that population is 0.9686 against the fuzzy arm's 0.9988. The mechanism agreement rate, pre-registered precisely to tell genuine practical equivalence apart from structural coincidence, came out at 0.9698, which is exact coverage plus the fuzzy abstention rate, $0.9686 + 0.0012$, to four decimal places. The diagnostic written to detect the structural case turned out to compute it.

4.3 The general form

The containment argument is about one pair. The trap is wider, in two steps of different reach.

Where the history names one author, a precision-first rule cannot win. This holds for any pair of mechanisms that assigns an author drawn from the block's own history, whatever the machinery inside them. If every training record in the block belongs to one author, any mechanism returning an author from a non-empty candidate pool returns that author, and the rule returns the same author whenever it fires. Its only available difference is to abstain, and abstention scores as incorrect, so the rule can lose there and can tie there, and cannot win.

Where the history is written one way, the rule cannot win either, for mechanisms that read surface forms only. If the block's training records carry a single distinct surface form, two mechanisms that take their evidence from surface forms and share a tie-break return the same author on every test mention both answer. A rule that abstains on unseen forms therefore cannot win, and it loses exactly where a test mention is written in a form the history never shows. A mechanism that also reads co-authors, venues or affiliations is outside this step, because it can separate two people who share one written form.

Both are visible in the data. On the evaluation split, in the blocks whose history names one author the rule is right and the vote wrong on 0 held-out signatures and the reverse holds on 3,029, every one of them a rule abstention. In the blocks whose history is written one way the counts are 0 and 696, again all abstentions, on the 835 held-out signatures there whose form the history never shows. The calibration-split probe of Section 5 finds the same two lines of zeros for all three candidate pairs (`phase2c_pair_regimes.csv`). A two-sided contest is possible only where the history is both impure and surface-varied, which is 42.2% of held-out signatures on the evaluation split.

The containment intuition is not new: the blocking literature already observes that key-based blocking is a special case of nearest-neighbour blocking under a binary similarity metric [Zhang et al., 2020]. What we add is the consequence drawn from it. Once one mechanism contains the other, one of the two verdicts is unreachable, so the comparison cannot inform the choice it was run to settle. The result is practical rather than deep, and the trap is easy to fall into: anyone comparing a string-matching baseline against a retrieval or clustering method over the same recorded history is inside the family. The check is cheap. Before running the comparison, count the items on which mechanism A is correct and mechanism B is not. A zero, or a zero confined to a stratum, was produced by the design and the data had no say.

5 A pair either side can win

5.1 Three candidates, and why a k-nearest vote

The repair cannot be a different similarity function, because the containment holds for every function that reserves its maximum for identical strings. Mechanism B has to stop being a single-nearest retriever. We probed three ways of doing that on the calibration split, using the 2,450 of its blocks that have at least two training signatures and one held-out signature, 35,732 held-out signatures in all, with mechanism A unchanged in all three pairs. Section 6.4 records that this was a second use of a split the frozen text had reserved for a single use. The *radius vote* pools every training record whose form is within similarity r of the test form and takes the most frequent author in the pool. The *normalized-key vote* pools every training record whose form carries the same ASCII-folded, lowercased, letters-only key as the test form. The *k-nearest vote* pools the nearest training records by form similarity, whole forms at a time, until at least k records are pooled, and votes.

All three are two-sided in the probe. We adopted the k-nearest vote at $k = 5$, most of all because it gives the most balanced contest: 546 signatures where the rule is right and the vote wrong against 744 the other way, with pooled accuracies 0.8234 and 0.8290. Rank also beats radius on this substrate, because every name in a block shares a surname, so absolute similarity has no natural scale and a fixed radius means different things in different blocks; the radius family confirms it, with the rule-side count running from 1,461 at $r = 0.70$ down to 0 at $r = 0.95$, so the whole finding would have been a function of r. And the k-nearest vote never abstains, which puts abstention entirely on the rule's side and separates the two arms along the precision-against-coverage axis the paper is about.

5.2 The calibration trap

Read the accuracy column of the same probe and a second problem appears. The k-nearest vote's pooled accuracy falls monotonically as k grows: 0.8390, 0.8354, 0.8290 and 0.8188 at k of 2, 3, 5 and 10. The radius vote's accuracy rises monotonically as r grows, reaching 0.8239 at $r = 0.95$, where its rule-side count is 0. In the radius family the accuracy-maximizing value on the grid is therefore $r = 0.95$, where the vote has shrunk to the rule's own evidence and the pair is nested again. In the k family it is $k = 2$, which is still two-sided at 181 against 736, but it is the grid point nearest $k = 1$, the single-nearest retriever of Section 4, and accuracy rises all the way toward it. Our original pre-registration calibrated the retriever's one parameter by picking the value with the highest pooled accuracy on the calibration split. Applied unchanged, that protocol selects a nested pair in one family and the grid point closest to one in the other. We therefore voided it. k is fixed by definition at 5, disclosed as a calibration-split choice made for contest balance rather than for accuracy, with a mandatory sensitivity curve over k in $\{2, 3, 5, 10\}$ reported beside every outcome. The general warning is worth more than the fix, and Section 8.2 states it.

6 Experimental design and pre-registration

6.1 Data, blocking and splits

The substrate is S2AND [Subramanian et al., 2021]. A *signature* is one author mention on one paper, carrying a name string and context fields. Signatures are grouped into *blocks* by a normalized name key, frozen before any measurement, and ground truth assigns each signature to a real author. We use seven of the eight constituent datasets: aminer, arnetminer, inspire, kisti, pubmed, qian and zbmath. Medline is excluded because the release ships no clusters file for it.

Table 1 accounts for every signature from the corpus down to the analysis population. Blocks with a single signature have no history to match against, so they sit outside the live set. Live blocks are split at block level with seed 20260918 into a 15% calibration split and an 85% evaluation split; the calibration split fixes the band edges and the cutoff of the mechanism we later retired, was used again for the design probe of Section 5.1 and for the regression check of Section 7.1, and supplies no reported hypothesis number. Inside each evaluation block, signatures are assigned to a training half and a test half (we keep the pre-registration's word, though the split is 70 to 30) with the same seed, by $n_{\text{train}} = \max(1, \min(n - 1, \lfloor 0.7n \rfloor))$. No test signature is visible to any signal, band edge, cutoff or mechanism.

The primary analysis population is evaluation blocks with at least two training signatures and at least one test signature, that is blocks of three or more signatures. Below that, consistency is 1.0 by construction on a one-signature history and surface variability is undefined, so neither signal carries information, and blocks of exactly two are a named and counted exclusion used to

decide nothing. A cell, band or dataset holding fewer than 30 blocks is printed with its count, marked under-powered and interpreted nowhere; a dataset qualifies for the generalization rule if it contributes at least 100 blocks.

Table 1: Population accounting, from the corpus to the primary analysis population. Both exclusions are named and counted rather than dropped. Source: `reports/phase2e_boundary.md`, generated by `phase2e_run.py`.

Stratum	Blocks	Signatures	Share of signatures
All signatures	26,939	766,133	100.00%
Singleton blocks, outside the live set	7,955	7,955	1.04%
Live blocks (2 or more signatures)	18,984	758,178	98.96%
Calibration split (15%)	2,848	116,308	15.18%
Evaluation split (85%)	16,136	641,870	83.78%
Primary analysis population	13,902	637,402	83.20%
Block-size-2 stratum, named exclusion	2,234	4,468	0.58%

6.2 The two signals and their bands

Both signals are computed from the training half of each block only, for the reason Section 3.1 gives.

ADS, the consistency signal, is the share of a block's training signatures belonging to its single most frequent author. It runs on $(0, 1]$ and equals 1.0 when the history names one person. Its bands are fixed by the definition of purity rather than read off a distribution: $[0, 0.5)$, $[0.5, 0.7)$, $[0.7, 0.9)$, $[0.9, 1.0)$ and exactly 1.0.

SV, the surface-variability signal, is the expected normalized Levenshtein distance between two training signatures of the block drawn at random without replacement, computed on the frozen surface form and weighted by signature. Its scale is bounded well below 1 by construction, because every name in a block shares a surname, so fixed bands on the nominal scale would be meaningless. SV = 0, a history written one way, is structural and is its own band; the other three are the terciles of the strictly positive values, with cut points taken from the calibration split at 0.1026 and 0.1951 and committed before any evaluation number existed.

Two controls travel with SV throughout, because the largest single source of within-block name variation in bibliographic data is forename initialization, the variable Kim and Kim [2020] manipulate. **SV_resid** is the same function applied to an initialization-canonical form, in which the first name and each middle token are reduced to their first letter, so that "John Smith" and "J. Smith" become the identical string and initialization contributes exactly zero, leaving spelling, diacritics, hyphenation, transliteration, middle-name presence and suffixes. Its band edges, calibrated the same way, are 0.0368 and 0.0824. **INIT** is the share of a block's training signatures whose given-name field is an initial only, banded at INIT = 0, $0 <$ INIT < 0.5 and INIT ≥ 0.5, with the SV analysis repeated inside each band. All four edges are printed to four decimals here and carried at full precision in `phase2b_constants.py`, the frozen record.

Both axes were checked alive and separable before any mechanism was built, as a hard gate on proceeding at all. Beyond the correlation quoted in Section 3.2, each measure tracks its S2AND counterpart strongly and the other weakly: Spearman -0.818 for ADS against homonymity and $+0.743$ for SV against synonymity, against cross terms of $+0.18$ and $+0.06$.

6.3 Mechanisms, metric and the winner

Mechanism A, the exact-evidence rule. If the test signature's surface form occurs among the training forms, assign the most frequent author among training signatures carrying that exact form. Otherwise abstain. No parameter.

Mechanism B, the k-nearest vote of Section 5.1, with $k = 5$. The pool is counted in signatures, not forms, and grows whole forms at a time; if the block holds fewer than k training signatures the pool is the whole block. Votes are weighted by signature. B does not abstain. Both mechanisms use the same frozen tie-break chain, most frequent author inside the pooled evidence, then most frequent author inside the block, then the lexicographically smallest author identifier, because different tie-breaks would turn part of the winner contest into a comparison of tie-break rules.

Metric. Whole-set accuracy over the block's test signatures, with abstention counted as incorrect, and coverage reported separately for each mechanism wherever accuracy is reported, because the asymmetry that the rule abstains and the vote does not is a result rather than a nuisance. Author name disambiguation is normally evaluated with B-cubed, pairwise F, cluster F or the K-metric [Amigó et al., 2009, Kim, 2019], and we use none of them: those metrics score a partition against a reference partition, which is right when the task is to discover clusters, whereas the task here is to assign each held-out mention to one author already present in the history, so the reference is a label per item. The cost is comparability, since no number here can be placed beside a published B-cubed F1 for the same block. We accept it because neither mechanism is offered as a disambiguation system and because the winner construct needs a per-item outcome it can resample.

Winner. Per block, a paired bootstrap on the rule's accuracy minus the vote's, resampling the block's test signatures with replacement, $B = 2000$ draws, both mechanisms scored on the same resample, read as a 95% percentile interval. Per band and per grid cell we pool the band's test signatures and bootstrap by resampling *blocks* with replacement, because the block is the independent unit. Four mutually exclusive verdicts against a margin δ: *rule wins* if the interval's lower bound exceeds $+\delta$, *k-NN wins* if its upper bound falls below $-\delta$, *tie* if the interval lies entirely inside $[-\delta, +\delta]$, and *undecided* otherwise, where *k-NN* abbreviates the k-nearest vote throughout. The primary margin is $\delta = 0.05$, five accuracy points, with $\delta = 0.02$ and $\delta = 0.03$ pre-registered as a robustness curve. An outcome holding at one margin and flipping at another is margin-dependent and is treated as Partial, never Supported, and the same rule applies across the k curve. And Inconclusive is evaluated first: if a hypothesis falls short of its pre-registered power floor, the outcome is Inconclusive whatever the other rows would have said, and a set of bands that are all tie or all undecided is never read as support.

6.4 Pre-registration and its amendment

The text of this subsection was written during the amendment pass, before the run it defends was executed, and locked in the project plan at that point. It is reproduced here as written.

> We pre-registered the hypotheses, the winner definition, the analysis bands, and the falsification table before computing any evaluation metric, and we froze the calibration constants in a separate commit before any evaluation number existed. The pre-registration and every subsequent amendment are in the public repository with full history.
>
> We amended the pre-registration once, after two exploratory runs, and we state that plainly rather than presenting the amended version as the original. The amendment

was forced by two structural facts about the mechanisms, not by an unwelcome effect size. First, when a test name is present in a block's history it sits at the maximum of any similarity function, so a single-nearest retriever returns whatever exact match returns; exact match is therefore a strict sub-case of single-nearest retrieval, and the two cannot contest a winner. Second, a block whose history names one author returns that author from any non-empty candidate pool, so at maximum consistency the only move a mechanism can make is to abstain. The first fact required replacing the second mechanism with a k-nearest vote that does not reduce to exact match. The second required scoping the consistency hypothesis to whether the choice is decidable at all, rather than which mechanism wins at every consistency level.

Three properties keep the amendment honest. The original hypotheses and the conditions of the runs that motivated the change remain in the pre-registration, unedited and dated. No accuracy figure for the k-nearest mechanism existed at the time the amended hypotheses were written; the amendment predates the run it governs. And the one component of the amended consistency hypothesis that is guaranteed by construction, the decline in decided winners as consistency approaches its maximum, is labeled as structural wherever it appears; the empirical content we test lies below that maximum, in the region where the outcome is not fixed in advance.

Eight points of record. The first three were checked at writing against the committed repository, and the third was extended at release. The next two were added after a simulated peer review, which the author ran on the manuscript with an AI assistant, found the locked text above too strong as written. The last three were added at release. All of the corrections work by disclosure and none by editing locked text. Times are Indian Standard Time, UTC+05:30.

What "amended once" means. After its freeze commit the pre-registration file changed three times. Two of the changes are dated appends, made in the same pass within half an hour of each other and both before the run they govern. The first is the amendment, which replaced the second mechanism and rewrote the two selection hypotheses. The second is an addendum replacing one clause's measure of which signal drives decidability, because that measure was an order statistic over a grid with five rows and four columns and so favoured the axis with more bands, which was the axis the hypothesis needed to win. One amendment event in two documents, and the addendum tightens a rule rather than relaxing one. The third change came first and is the subject of the third correction below.

What the two runs before the amendment were. The first was the pre-registered run, executing the frozen text exactly, and it is what measured the degeneracy in Section 4.2. The second was exploratory by construction: a design probe confined to the calibration split, producing no evaluation-split number and evaluating no hypothesis.

The ordering claim, against commit timestamps. The pre-registration was committed before the run that executed it, and the calibration constants in an earlier commit of their own before any evaluation-split accuracy existed. The amendment was committed just under two hours before the commit that records the run it governs, and the addendum about eighty minutes before it.[1] The run itself began sooner. File-system times on the author's machine, which are not repository evidence and are set out in `phase2/reports/phase2_9b_report.md`, put the start of the first run about seventeen minutes after the addendum and its cached evaluation-split predictions at about twenty-three. That run stopped on a programming error in its reporting stage. The analysis script

[1] Commits, in order: `8869b82` freezes the pre-registration; `a41a67a` inserts two rules into it nine hours later, before any measurement; `5e13e38` freezes the calibration constants; `3d86959` is the first run; `c20239a` is the design probe; `5f452ad` and `3e79a48` are the amendment and its addendum; `9e3d921` is the run reported here.

was then edited for another forty minutes, the recorded edits being a crash fix and a floating-point tolerance, and every later run, including the one whose tables are reported, reused the cached predictions. The analysis code was therefore finished after evaluation-split predictions existed. The claim that survives is the narrower one: the hypotheses, the decision rules and every constant were committed before any of those predictions existed. The reported run opens with the regression check of Section 7.1, which reproduces calibration-split counts frozen before the amendment was written.

First correction: what existed when the amendment was written. The quote says that no accuracy figure for the k-nearest mechanism existed when the amended hypotheses were written. That holds of the evaluation split and not without the qualifier, and since the quote is reproduced unedited the correction belongs here. Commit `c20239a`, at 12:09:23 on 2026-09-18 and twenty-seven minutes before the amendment, holds `phase2c_pair_twosidedness.csv` and `phase2c_pair_regimes.csv`: between them the k-nearest vote's pooled calibration-split accuracy at k of 2, 3, 5 and 10, namely 0.8390, 0.8354, 0.8290 and 0.8188, and its per-cell gap against the rule over the same five by four ADS-by-SV grid the amended H2 and H3 are written over. Those gaps already carry the compression-and-sign-change shape the amended hypotheses go on to predict, at $+0.0576$ in $[0, 0.5)$ crossed with SV-high against -0.0526 at ADS $= 1.0$ crossed with SV $= 0$, so the amended hypotheses were written with the calibration-split grid in view. What makes that defensible is not the absence of information but the bar on using the calibration split for any reported hypothesis number, with the evaluation split untouched: Amendment 1's wording is that no mechanism B prediction on the evaluation split had been computed, scored or printed, and that scoped claim is the one that survives being checked.

Second correction: what the ordering evidence is not. All of the above is author-controlled git metadata. Commit dates can be set and history can be rewritten, so the record establishing the ordering is attested by the party it exculpates, and a reader is entitled to weigh it on those terms. One control is not author-controlled in the same way. The reported run opens by reproducing calibration-split counts that were frozen before the amendment was written, and it halts if they differ, so every number in this paper comes from a run tied to a state of the repository that predates the hypotheses it tests. The repository was pushed and tagged at release, as `v2.0.0`, so that the commits named here resolve for a reader. A push in November proves nothing about the order of events in September, and we do not offer it as if it did.

Third correction: the freeze was followed by an insertion. Nine hours after the freeze commit, commit `a41a67a` inserted two rules into the frozen text, with a dated note at the head of the file saying so. One makes $\delta = 0.05$ the primary margin and requires the curve at 0.02 and 0.03. The other fixes the order of evaluation, under which Inconclusive is read first, and that is the rule that returns both selection outcomes in Section 7.4. Both tighten the table. Without the second, a run in which every band was a tie or undecided could have met the original H2's Supported row, which is the gap the inserted rule closes. The rules were written outside the assistant session that drafted the pre-registration, at the review held between that pass and the next, and were inserted verbatim. Commit order and local file times both put the insertion before any calibration or evaluation number; the evidence is set out in `phase2/reports/phase2_9a_report.md` and is author-controlled like the rest.

Fourth correction: the calibration split was used more than once. The frozen text says the calibration split is used exactly once, for the retired cutoff and the band edges. The design probe used it a second time, to choose the replacement pair and k, and the reported run a third, for its regression check. Amendment 1 discloses the second use and describes it as the kind of choice the split exists for, which is more than the frozen sentence says. What held throughout is the other half of that sentence: no reported hypothesis number comes from the calibration split.

Fifth correction: what "any similarity function" needs. The quote says a test name present in the history sits at the maximum of any similarity function, so a single-nearest retriever returns what exact match returns. The conclusion needs the maximum to belong to identical strings alone. Section 4.1 states that condition, shows it holds for both functions used here, and names common ones for which it fails.

7 Results

Every number below comes from the evaluation split, from the 13,902 blocks and 197,479 test signatures of the primary analysis population, at $k = 5$ in the Levenshtein arm and at $\delta = 0.05$ unless stated. The three outcomes are read off the falsification table frozen before the run, in the order it fixes. Anything reported beside an outcome is substance, and is labelled as substance.

7.1 The pair is two-sided now

The run opens with a regression check on the calibration split, executed before any evaluation signature is read. The k-nearest vote is implemented independently of the design probe of Section 5.1, and the check asserts its two-sidedness counts against the frozen probe table. All four k values reproduce exactly, including 546 against 744 at $k = 5$, with the rule's accuracy 0.8234 identical across k as it must be, and the run halts if they do not.

On the evaluation split the amended pair is two-sided at every k on the mandatory curve (Table 2). At $k = 5$ there are 3,025 held-out signatures on which the rule is right and the vote wrong, against 4,012 the other way; the Jaro-Winkler arm gives 2,946 against 4,002. The retired pair measured zero on the rule's side across 199,713 signatures. A winner now exists to be predicted.

Table 2: Two-sidedness on held-out data, for the retired pair and for the amended pair across the mandatory k curve and both metric families. The retired pair's zero was counted over the whole evaluation split, 16,136 blocks and 199,713 signatures; every accuracy and gap is measured on the primary analysis population. The retired pair's 3,876 is fixed by the containment rather than measured against it, being the accuracy gap times the test-set size. Sources: `phase2e_twosidedness_eval.csv`; `reports/phase2b_boundary.md` and `phase2b_accuracy_by_population.csv` for the retired row.

Pair	k	A right, B wrong	B right, A wrong	Rule acc.	B acc.	Pooled gap
Exact vs. fuzzy (retired)	1	**0**	3,876	0.8244	0.8441	−0.0196
Rule vs. k-NN, Levenshtein	2	987	3,990	0.8244	0.8396	−0.0152
Rule vs. k-NN, Levenshtein	3	1,724	4,004	0.8244	0.8360	−0.0115
Rule vs. k-NN, Levenshtein	5	3,025	4,012	0.8244	0.8294	−0.0050
Rule vs. k-NN, Levenshtein	10	5,176	4,040	0.8244	0.8187	+0.0058
Rule vs. k-NN, Jaro-Winkler	5	2,946	4,002	0.8244	0.8298	−0.0053

7.2 The consistency signal is a real difficulty signal

H1 asks whether train-only ADS tracks each mechanism's own held-out accuracy. It is a construct-validity check and not a contribution: that disambiguation accuracy falls as homonymity rises is already visible at dataset level in Kim and Kim's homonym-centric datasets and in the S2AND

paper, and what H1 confirms is that our per-block statistic behaves like the established dataset-level quantity.

The Spearman correlation of train-only ADS with held-out accuracy is 0.5069, 95% block-bootstrap interval $[0.4919, 0.5224]$, for the rule, and 0.8298, $[0.8190, 0.8401]$, for the vote. Both intervals sit well clear of zero, and the same holds in the pooled non-inspire analysis, at 0.5632 and 0.7659, and in all four qualifying non-inspire datasets. Under the frozen rule, **H1 is Supported**. Across the first four ADS bands mean per-block rule accuracy rises 0.4991, 0.6758, 0.8389, 0.9418 and the vote's rises 0.4088, 0.6296, 0.8165, 0.9465. In the top band the vote rises again to 0.9889 and the rule falls back to 0.9234, for the reason in the caption (Figure 1). Measuring your data at design time does tell you how hard the problem will be. The rest of this section is about what it does not tell you.

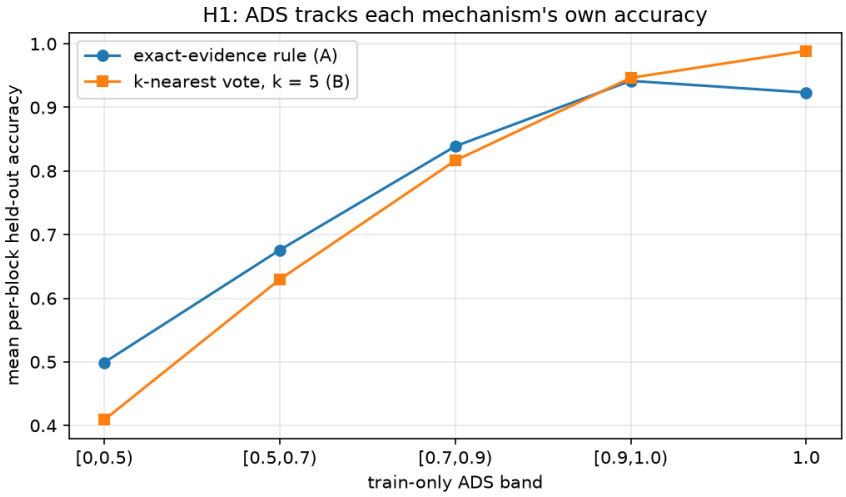


Figure 1: Mean per-block held-out accuracy by train-only ADS band, for both mechanisms. The consistency signal predicts difficulty for each mechanism separately. In the top band the history names one author, so the rule's only possible departure from the vote is to abstain (Section 4.3), which is why it sits below the vote there. Written by `phase2e_run.py` from the band means of `phase2e_blocks_scored.csv.gz`.

7.3 Head to head

Pooled, the rule scores 0.8244 and the vote 0.8294, a gap of -0.0050. Half an accuracy point. Rule coverage is 0.9686 against the vote's 1.0, so abstention sits entirely on the rule's side by design and is scored as incorrect throughout, and the two mechanisms return the identical prediction on 94.93% of test signatures.

Scoring abstention as incorrect is the right primary metric for a choice between deployable mechanisms, and it is not the number the precision-first framing invites, so we print that one too. The rule fires on 96.86% of held-out signatures and is right on 0.8512 of those. The vote is right on 0.8359 of the same signatures, so where the rule commits it is the more accurate arm by 0.0153, with a block-bootstrap interval of $[+0.0142, +0.0165]$, and it pays for that with 3.14% abstention. The like-for-like margin is positive in every dataset.

Neither mechanism can pass a ceiling the data sets for both. Both assign an author drawn from the block's own training half, so a test signature whose true author never appears in that

half is an automatic error for either arm. Pooled, 4.14% of held-out signatures are of that kind, so nothing in this mechanism family can exceed 0.9586 here, and Table 3 gives the ceiling per dataset. It is a shared floor subtracted from both arms, so it moves each accuracy and barely moves their difference, and it is not what separates the datasets: aminer's ceiling is 0.8435 against inspire's 0.9964, a difference of 0.1529, while their rule accuracies differ by 0.5406 and remain 0.4684 apart when each is normalized to its own ceiling, so the ceiling accounts for about 13% of the spread and the rest is the blocks being harder.

Table 3 breaks the comparison down and the picture is not uniform. inspire carries 11,110 of the 13,902 blocks, so the pooled number is largely an inspire number, and pooled over the six non-inspire datasets the rule is ahead by 0.0221; the ceiling column says what makes inspire different, at 0.36% of its held-out signatures naming an author its block's history never shows against 13.20% pooled over the other six. kisti carries the largest gap of any qualifying dataset, +0.1131 to the rule, on the lowest agreement rate in the table, 0.7826, which makes it the clearest single instance of the precision-against-coverage trade. Table 2 carries one more comparison. The retired single-nearest retriever, at 0.8441, is more accurate pooled than either mechanism of the amended pair. It is the rule with a fallback wherever the rule abstains, so it is right everywhere the rule is right, and a designer who wants accuracy alone needs no measurement to prefer it to the rule. Against the vote its lead is 0.0146, inside the margin. The vote's accuracy falls monotonically in k, at 0.8396, 0.8360, 0.8294 and 0.8187, which is the calibration trap of Section 5.2 on the evaluation split.

Table 3: Held-out accuracy, coverage and agreement per dataset, with inspire alone and pooled non-inspire. *Ceiling* is one minus the share of test signatures whose true author is absent from the block's training half: an automatic error for both arms, so no mechanism in this family can exceed it. Datasets below 100 blocks do not qualify for the generalization rule and decide nothing. Sources: `phase2e_accuracy_by_population.csv`; `phase2_7b_accuracy_accounting.csv` for the ceiling.

Population	Blocks	Test sigs.	Ceiling	Rule	k-NN	Gap	Rule cov.	Agree.
All (primary)	13,902	197,479	0.9586	0.8244	0.8294	−0.0050	0.9686	0.9493
aminer	842	40,006	0.8435	0.4076	0.4038	+0.0038	0.9800	0.9620
arnetminer	82	1,620	0.8488	0.5093	0.5099	−0.0006	0.9796	0.9716
inspire	11,110	139,369	0.9964	0.9482	0.9645	−0.0163	0.9705	0.9612
kisti	711	10,147	0.9049	0.8466	0.7334	+0.1131	0.9218	0.7826
pubmed	34	803	0.9888	0.8020	0.5554	+0.2466	0.9489	0.6650
qian	276	1,696	0.9517	0.6079	0.5896	+0.0183	0.9876	0.9629
zbmath	847	3,838	0.9713	0.8494	0.9125	−0.0631	0.8953	0.8682
Pooled non-inspire	2,792	58,110	0.8680	0.5276	0.5054	+0.0221	0.9640	0.9206

7.4 What the frozen table returned

Per-block verdicts at the primary margin: 191 rule wins, 309 k-NN wins, 10,172 ties and 3,230 undecided. Five hundred of 13,902 blocks carry a decided verdict, and they are not spread evenly over block sizes. A block with one test signature has a resampling distribution that is a point mass, so its interval is a single point and *undecided* is unreachable for it by construction rather than by evidence. The size-1 stratum holds 1,309 blocks, at 14 rule wins, 165 k-NN wins, 1,130 ties and, necessarily, zero undecided, and the size-2 stratum holds 2,159, at 10, 89, 1,739 and 321. Together they supply 278 of the 500 decided verdicts and 2,869 of the 10,172 demonstrated ties.

We therefore recomputed everything the per-block verdict feeds on the 10,434 blocks carrying at least three test signatures, 75.05% of the blocks and 97.15% of the test signatures. This is a sensitivity analysis; the frozen primary analysis is the all-blocks column above, unchanged. No outcome moves: zero ADS bands, zero SV bands and zero of the twenty grid cells carry a decided pooled verdict at $\delta = 0.05$ under the restriction, so the power floor still binds. What moves is composition. The decided rate falls from 0.0360 to 0.0213, the tie share from 0.7317 to 0.6999 and the undecided share rises from 0.2323 to 0.2788, which is the degeneracy being removed rather than evidence changing, and the per-block margin reverses, 191 rule wins against 309 becoming 167 against 55.

H2 is Inconclusive and H3 is Inconclusive, both at the primary margin, for the same reason. Inconclusive is evaluated first and neither hypothesis reaches its power floor. No ADS band and no SV band has a decided pooled verdict at $\delta = 0.05$: every marginal interval on both axes lies inside the margin, so every marginal pooled verdict is a demonstrated tie. No cell of the five by four grid is decided either: eighteen of the twenty are ties and the other two are undecided, those being $[0, 0.5)$ crossed with SV-high, the most contested cell in the grid at a decided rate of 0.1971 over 553 blocks, and ADS = 1.0 crossed with SV = 0, the largest at 3,922 blocks. H2 requires two decided ADS bands and two decided SV bands and has none of either; H3 requires two decided SV bands inside a single ADS band and has none. The pre-registered margin is what made these outcomes Inconclusive, which means the frozen rules declined to decide rather than that the effects below the margin are absent, and both floors were pre-listed as permitted outcomes.

The margin sensitivity curve was pre-registered for the same reason (Table 6). H2 is Inconclusive at all three margins; H3 is Inconclusive at 0.05 and 0.03 and Partial at 0.02, which under the frozen rule makes it margin-dependent. The power picture is in Figure 2: ties dominate at every test-set size, and undecided grows with size to roughly a third of blocks between nine and sixty-four test signatures.

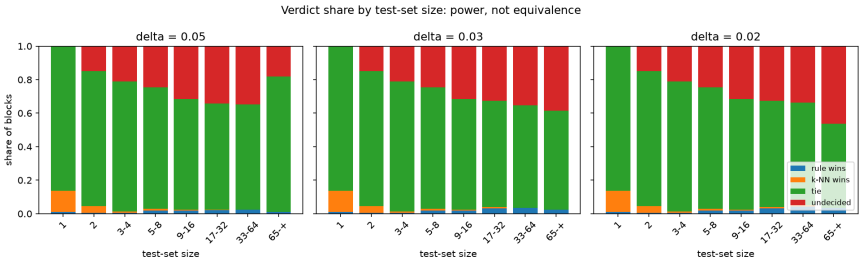


Figure 2: Verdict share against test-set size, with tie and undecided kept apart, because a demonstrated tie is not the same as insufficient evidence. The size-1 stratum cannot produce an undecided verdict by construction, so its tie bar is not evidence of equivalence; Section 7.4 gives that stratum's counts and the sensitivity analysis that removes it. Written by `phase2e_run.py` from `phase2e_power_by_size.csv`.

7.5 Gap compression on both axes

What follows is substance reported beside the verdicts. None of it is a hypothesis outcome and none of it is offered in place of one.

The pooled gap changes sign along both axes and is compressed along both (Table 4, Figure 3). Across ADS bands it runs +0.0334, +0.0215, +0.0123, −0.0028, −0.0278, from the rule ahead where the history is most impure to the vote ahead where it names one person. Across SV bands it runs

−0.0388, −0.0144, −0.0115, +0.0189, from the vote ahead where the history is written one way to the rule ahead where it is most varied. The largest magnitude anywhere on either axis is 0.0388. At $\delta = 0.02$ the ADS ordering becomes a decided marginal reversal, rule wins in $[0, 0.5)$ against k-NN wins at ADS = 1.0. Below ADS = 1.0 there is no SV-driven winner reversal at any margin in the direction-by-SV grid, the strongest structure below the top band there being a tie-crossing, at $\delta = 0.03$ in one band and at $\delta = 0.02$ in two. Substantively the shape is the pre-registered gap-compression one, which is Kim and Kim's dataset-level finding reproduced at block level with surface variation measured rather than manipulated.

Robustness, gathered in one place. SV_resid, which removes forename initialization by construction, gives the same shape at smaller magnitude and with no reversal, −0.0124, −0.0025, −0.0025, +0.0130, so the surface effect is more than initialization and small in either form. The Jaro-Winkler arm reproduces the SV sequence at −0.0388, −0.0146, −0.0116, +0.0181 with the same verdicts. The compact regression of the per-block gap agrees in sign, with coefficients −0.1889 on ADS (s.e. 0.0076), +0.0548 on SV (s.e. 0.0212) and +0.0208 on log test-set size (s.e. 0.0018); it summarizes the grid and decides nothing. Inside the INIT bands the shape survives as a tie-crossing at INIT = 0 and at INIT ≥ 0.5, and flattens to compression only in between, at $\delta = 0.05$.

Two winner reversals exist at $\delta = 0.02$ in the pooled grids we report at $\delta = 0.05$, and they belong here rather than in an appendix a reader has to find. Across ADS bands at fixed SV-high the sequence is rule wins, rule wins, tie, tie, k-NN wins, which is three decided bands of opposite identity and a full winner reversal under our own definition. Inside INIT = 0 the SV sequence carries k-NN wins at SV = 0 against rule wins at SV-high, a reversal as well. Both favour the reading that structure exists below the pre-registered margin, and Appendix A gives the winner structure for all three grids at all three margins so that no margin is reported only where it helps us.

Table 4: Marginal pooled gap (rule minus k-NN) and pooled verdict by ADS band and by SV band, at all three pre-registered margins, with the decided-winner rate at the primary margin. No band is decided at $\delta = 0.05$, which is the shortfall against H2's power floor. H3's floor is a different condition, counting ADS bands that hold two decided SV bands; the count there is also zero, and the grid behind it is Figure 3. Sources: `phase2e_pooled_ads.csv`, `phase2e_pooled_sv.csv`.

Band	Blocks	Test sigs.	Gap	δ=0.05	δ=0.03	δ=0.02	Decided
By consistency (ADS)							
$[0, 0.5)$	1,071	47,167	+0.0334	tie	undecided	rule wins	0.1083
$[0.5, 0.7)$	1,101	16,079	+0.0215	tie	tie	undecided	0.0618
$[0.7, 0.9)$	743	12,660	+0.0123	tie	tie	tie	0.0256
$[0.9, 1.0)$	450	12,683	−0.0028	tie	tie	tie	0.0089
1.0	10,537	108,890	−0.0278	tie	tie	k-NN wins	0.0278
By surface variability (SV)							
SV = 0	4,517	17,960	−0.0388	tie	k-NN wins	k-NN wins	0.0354
SV low	3,032	56,408	−0.0144	tie	tie	tie	0.0142
SV mid	3,218	59,433	−0.0115	tie	tie	tie	0.0177
SV high	3,135	63,678	+0.0189	tie	tie	undecided	0.0766

7.6 Which signal drives decidability

The amended pre-registration carried one clause comparing the two signals directly, asking which drives whether the mechanism choice is decidable at all. It required consistency to be the stronger

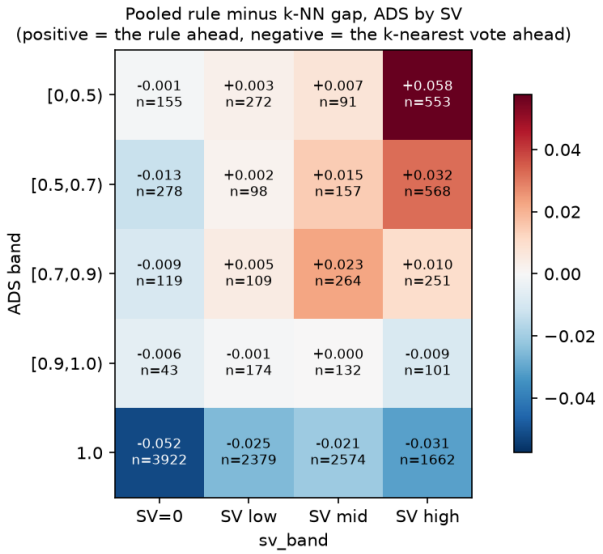


Figure 3: Pooled gap over the frozen five by four grid of ADS bands crossed with SV bands. The empirical half of the result in one image: compression on both axes, a sign change on both, and no cell far from zero. Written by `phase2e_run.py` from `phase2e_pooled_cells_sv.csv`.

driver under two measures at once: a variance decomposition over the eligible cells of the frozen grid, and an averaged range across each axis. Consistency loses under both (Table 5). The decomposition gives $\eta^2_{\text{ADS}} = 0.1749$ against $\eta^2_{\text{SV}} = 0.3837$, and the averaged ranges give 0.0707 against 0.0755, so under the addendum's own rule the clause *failed* rather than splitting. It fails at every margin on the robustness curve, and across the k curve it fails at $k = 3$, 5 and 10 and splits at $k = 2$, so consistency wins both required measures at no point. Removing the structural ADS = 1.0 row widens the gap rather than closing it, at 0.1755 against 0.4594; that restriction is a labelled diagnostic and decides nothing, as is the block-weighted variant, which gives 0.3653 against 0.3765 at the primary margin and turns the other way at the two smaller ones, 0.3972 against 0.3443 and 0.4114 against 0.3418.

Two point statistics do not say how firm a comparison between them is, so we bootstrapped their difference by resampling blocks with replacement, $B = 2000$ draws, at the seed scheme the frozen analysis already uses. At $k = 5$, $\eta^2_{\text{SV}} - \eta^2_{\text{ADS}}$ is $+0.2088$ with a 95% interval of $[+0.0628, +0.3473]$, and surface variability is the larger in 99.9% of draws. The grid cells are not balanced on test-set size and decidability rises with it, so we also partialled log cell-mean test-set size out of the cell rates and repeated the decomposition on the residuals, giving $+0.1840$, $[+0.0299, +0.3277]$: the confound is present and is not load bearing. At $k = 2$ the difference is $+0.0474$, $[-0.2531, +0.3159]$, with surface variability the larger in only 57.0% of draws; the size-adjusted version there is $+0.2725$, $[+0.0093, +0.4691]$, which clears zero narrowly. The clause outcome is therefore stable across k and the evidence behind it is not: the point estimate points the same way at every k, while the direction is decisive at $k = 5$ and at $k = 2$ depends on the adjustment, which is also the one point where the clause was already reported as split. This is the clause the amended consistency hypothesis needed consistency to win. It does not decide H2, because the power floor is read first, and it points against the roles we had assigned the two signals.

Beside the clause sits the decidability gradient itself, plotted across ADS bands at fixed SV in `phase2/reports/figs/phase2e_decidability_gradient.png`. The marginal decided-winner

Table 5: Decidability drivers. The clause required consistency to lead on both measures; it leads on neither. The lower panel bootstraps the difference between the two decompositions by resampling blocks, $B = 2000$, and the size-adjusted row partials log cell-mean test-set size out of the cell rates first. Neither the bootstrap nor the two diagnostic rows decides anything. Sources: `phase2e_driver_comparison.csv`, `phase2_7b_driver_bootstrap.csv`.

Grid	δ	η^2_{ADS}	η^2_{SV}	Range ADS	Range SV	Clause
Full	0.05	0.1749	0.3837	0.0707	0.0755	failed
Full	0.03	0.1664	0.3787	0.0800	0.0901	failed
Full	0.02	0.1720	0.3783	0.0887	0.0953	failed
ADS < 1.0 (diagnostic)	0.05	0.1755	0.4594	0.0622	0.0875	decides nothing
Block-weighted (diagnostic)	0.05	0.3653	0.3765	n/a	n/a	decides nothing
Bootstrapped difference $\eta^2_{\text{SV}} - \eta^2_{\text{ADS}}$, 95% interval, and share of draws with SV larger						
$k = 5$, full grid	0.05	$+0.2088$	$[+0.0628, +0.3473]$	99.9%		
$k = 5$, size-adjusted	0.05	$+0.1840$	$[+0.0299, +0.3277]$	99.5%		
$k = 2$, full grid	0.05	$+0.0474$	$[-0.2531, +0.3159]$	57.0%		
$k = 2$, size-adjusted	0.05	$+0.2725$	$[+0.0093, +0.4691]$	97.8%		

rate runs 0.1083, 0.0618, 0.0256, 0.0089 across the first four ADS bands and then rises again to 0.0278 at ADS = 1.0. The amendment predicted it would approach zero there, on the structural ground that a block naming one author leaves the rule no move but abstention. The reasoning was right and the prediction was wrong: the rate rises, because in the blocks where the rule abstains often the vote wins by a wide and decidable margin, and the top band holds 10,537 of the 13,902 blocks. The clause requiring the gradient to be non-increasing across ADS holds in none of the four usable SV bands.

The rise is a small-block artifact, and the sensitivity analysis of Section 7.4 shows it. Restricted to blocks with at least three test signatures the marginal gradient runs 0.1096, 0.0443, 0.0254, 0.0089, 0.0070, which is monotone: the top-band rate falls from 0.0278 to 0.0070 and sits below the band beneath it. One of the two per-SV clause counts moves with it, from two of four usable SV bands to three of four, and the non-increasing count from none of four to one of four. None of this moves H2, which is decided on the power floor before any clause is read. On the frozen primary analysis the amendment's prediction failed, and on blocks large enough for the interval to carry information it holds.

Two pre-registered reporting obligations close the subsection. The clause requiring the top ADS band to be strictly lower than the bottom admits a per-SV-band and a marginal construction, and here they disagree, holding in two of four usable SV bands and holding marginally; we print both, and neither was picked after seeing which way it fell. And the non-trivial decidable region required below the top band exists, the $[0, 0.5)$ and $[0.5, 0.7)$ bands each holding thirty or more blocks with a decided per-block verdict.

7.7 Block size, the one signal whose own bands decide in the pool

The compact regression above carries a third term, log test-set size, at $+0.0208$ per log unit. Block size is the count of signatures in a block, readable from the blocking pass alone with no labels and nothing built, so it meets the same design-time test ADS and SV meet. We did not pre-register it, and nothing here is a pre-registered test: it is a supplementary analysis run on the committed per-block records after the frozen analysis was complete. We report it because a general claim

Table 6: The falsification table evaluated, with the fact that decided each outcome, plus the pre-registered margin and k sensitivity curves. Every outcome is read off a rule frozen before the run. Sources: `reports/phase2e_boundary.md`, `phase2e_margin_sensitivity.csv`, `phase2e_k_curve.csv`.

	Outcome	Deciding fact
H1	Supported	Spearman positive with interval excluding zero for both mechanisms, 0.5069 and 0.8298, and generalizing to pooled non-inspire and all four qualifying non-inspire datasets.
H2	Inconclusive	Below the power floor, which is evaluated first: zero ADS bands and zero SV bands with a decided pooled verdict at $\delta = 0.05$.
H3	Inconclusive	Below the power floor, which is evaluated first: zero ADS bands containing two decided SV bands.
Margin sensitivity		
$\delta = 0.05$	H2 Inconclusive, H3 Inconclusive	
$\delta = 0.03$	H2 Inconclusive, H3 Inconclusive	
$\delta = 0.02$	H2 Inconclusive, H3 Partial, so H3 is margin-dependent	
k sensitivity		
$k = 2$	Decidable region present; driver clause split	
$k = 3, 5, 10$	Decidable region present; driver clause failed	

about design-time signals cannot rest on the two we happened to choose.

Banded in octaves (Table 7), block size is the only design-time signal in this study whose own bands carry a decided verdict at the primary margin when all seven datasets are pooled. Blocks of three to four signatures give a pooled gap of -0.1145, $[-0.1286, -0.1007]$, k-NN wins; blocks of five to eight give -0.0712, $[-0.0807, -0.0614]$, k-NN wins. The first is more than twice the pre-registered margin, and the pooled gap is monotone in band order across all seven. Neither ADS nor SV produces one decided marginal band, and that absence is what carried H2 and H3 to Inconclusive. The initialization share, the third train-only quantity we compute, has no decided band of its own either, at gaps of $+0.0066$, $+0.0018$ and -0.0183. Crossed with SV in the pre-registered control, it does have two decided cells. Initial-heavy histories written one way give the vote -0.1071, $[-0.1237, -0.0912]$, with no signature on which the rule is right and the vote wrong, which is the abstention pattern again. Histories with no initials and high variability give the rule $+0.1363$, $[+0.1182, +0.1557]$, and that cell is one dataset: kisti supplies 3,781 of its 4,746 held-out signatures, and without kisti its gap is $+0.0052$, a tie. Section 7.8 takes kisti up directly. Read on its own this table is a counter-example to a flat claim that no design-time signal here is actionable, and it is stated here before it is qualified. Two measurements qualify it, both on the same records.

At fixed consistency the sign never changes. The axes correlate, block size running at Spearman -0.2515 with ADS and $+0.3976$ with SV, so this table has to be read at fixed ADS before it can be read as a statement about block size at all. Split at the structural top row, inside ADS $= 1.0$ the pooled gap is negative in all seven bands and runs -0.1342, -0.0899, -0.0661, -0.0413, -0.0271, -0.0136, -0.0076, shrinking monotonically toward zero; inside ADS < 1.0 it is positive in all seven and runs $+0.0594$, $+0.0370$, $+0.0854$, $+0.0759$, $+0.0428$, $+0.0302$, $+0.0118$. Neither stratum changes sign anywhere. What changes across the marginal bands is the mix: 90.0% of the blocks in the smallest band are at ADS $= 1.0$ against 43.8% in the largest, so the marginal reversal

is composition. Inside a stratum block size decides every band the same way, which is a tie-crossing in the frozen vocabulary of Appendix A and a winner reversal nowhere. Block size says how much the choice is worth; consistency has already said which way it points, and at fixed consistency block size never points anywhere else.

The decided bands are an abstention effect, not a resolution effect. The rule's coverage climbs monotonically with block size, at 0.8423, 0.8927, 0.9217, 0.9521, 0.9631, 0.9786 and 0.9827, and abstentions are scored as errors, which is where the small-block deficit comes from. On the signatures where the rule fires, its precision exceeds the vote's accuracy on those same signatures in all seven bands, including both decided ones: 0.9370 against 0.9230 in the smallest. Telling a designer to use the vote on small blocks is telling them to prefer the mechanism that is less often right where it commits, because the other declines to commit on about a sixth of the items there. That is a statement about tolerance for abstention, not a finding about which mechanism resolves a name better.

Block size is also what reconciles the pooled headline with the per-block count. Pooled over signatures the vote is ahead by 0.0050; per block the rule takes 191 wins against the vote's 309. Both are true and coverage separates them. The vote's per-block wins live in small blocks: the two smallest bands hold 263 of its 309, against 24 of the rule's 191, and above sixty-four signatures the vote wins no block at all. The signatures live in large blocks: the two largest bands are 2,809 of 13,902 blocks but 131,083 of 197,479 test signatures, so a one-block-one-vote tally and a one-signature-one-vote pool are counting almost disjoint majorities. Nor is the pooled gap carried by the bands that hold the signatures: the five bands with a negative gap contribute a net deficit of 1,466 signatures on 66,396, the two with a non-negative gap contribute +479 on 131,083, and the total of −987 over 197,479 is the pooled −0.0050. On the restricted population of Section 7.4 the pattern holds, with the per-block margin 167 rule wins against 55, the 3-4 band gone because those blocks hold one or two test signatures by construction, and the 5-8 band still decided at −0.0648.

What this licenses, and what it does not. Two different things get called a winner here. Block size predicts which mechanism has the higher whole-set accuracy on small blocks, at the primary margin, and that mechanism is the one that never abstains. It does not predict which mechanism resolves a name better. That edge is a mechanical consequence of the vote's full coverage against the rule's abstention, an asymmetry fixed in Section 6.3 before anything was measured, so nobody had to measure block size to learn it. This is the one place where the claim that no design-time signal here is actionable, stated flat, is too strong, so we do not state it flat. What the evidence supports is what we claim: pooled over the seven datasets, no design-time signal we examined tells a designer which mechanism resolves a name better, and the one signal whose bands do yield a decided verdict yields it through the rule's abstention rather than through a difference in matching quality.

7.8 Two datasets are not like the pool

The pre-registration requires every headline table in three forms: per dataset, inspire alone, and pooled over the other six. The frozen report gives the accuracy and H1 tables that way and the band tables only pooled. We computed the missing forms from the committed per-block records with the frozen bootstrap, in `phase2/reports/phase2_9b_report.md`. This meets a reporting obligation late. It is not a new test, and it changes no outcome, because the falsification table is read on the pooled population.

It does change the picture. Of the five datasets large enough to judge, three are ties at the primary margin and two are decided, in opposite directions and for different reasons. In kisti the rule is ahead by 0.1131, $[+0.1015, +0.1250]$, *rule wins*. The advantage sits where the pooled signs

Table 7: Pooled gap (rule minus k-NN) by block-size band on the primary analysis population, with the rule's coverage and its precision where it fires. The two decided bands are the only decided verdicts in any signal's own bands when all seven datasets are pooled; the last two columns are why they are an abstention effect. Bands are octaves, fitted to no outcome, and this analysis was not pre-registered. Sources: `phase2_7b_blocksize_pooled.csv`, `phase2_9a_matched_precision.csv`.

Band	Blocks	Test sigs.	Gap	95% interval	δ=0.05	Rule cov.	Rule prec.
3–4	2,227	3,145	−0.1145	[−0.1286, −0.1007]	k-NN wins	0.8423	0.9370
5–8	2,188	5,323	−0.0712	[−0.0807, −0.0614]	k-NN wins	0.8927	0.9529
9–16	2,181	8,675	−0.0350	[−0.0430, −0.0271]	tie	0.9217	0.9391
17–32	2,293	17,194	−0.0118	[−0.0169, −0.0065]	tie	0.9521	0.9424
33–64	2,204	32,059	−0.0069	[−0.0101, −0.0034]	tie	0.9631	0.9560
65–128	1,652	45,924	+0.0001	[−0.0027, +0.0031]	tie	0.9786	0.9599
129+	1,157	85,159	+0.0056	[+0.0037, +0.0077]	tie	0.9827	0.7194

k-NN accuracy on the signatures where the rule fires, in the same seven bands:
0.9230, 0.9388, 0.9130, 0.9172, 0.9375, 0.9456, 0.7076. The rule is ahead in all seven.

said it would: +0.1762 and +0.1100 in the two most impure ADS bands and +0.0806 and +0.1654 in the two most varied SV bands, each *rule wins*, and a tie where the history names one author or is written one way. It is a difference in matching. Across kisti the rule is right and the vote wrong on 1,219 held-out signatures and the reverse holds on 71, and on the signatures where the rule commits it is ahead by 0.1299. In zbmath the vote is ahead by 0.0631, [−0.0761, −0.0512], *k-NN wins*, and the reason is abstention. Its deficit sits in the one-author and one-form bands, at −0.0976 and −0.1137, where the rule can only lose by declining to answer, and its rule coverage, 0.8953, is the lowest of any qualifying dataset. Where the rule commits in zbmath it is still ahead, by 0.0221. zbmath's upper bound clears the margin by only 0.0012, which is narrow; at the two smaller margins the vote wins with room to spare. The one smaller dataset with a decided verdict, pubmed at 34 blocks, points the same way as kisti and is too small to qualify.

Pooled over the six non-inspire datasets, the grid has decided cells in both directions at the primary margin: *rule wins* in the two most impure ADS bands crossed with SV high, at +0.0629 and +0.1024, and *k-NN wins* at ADS = 1.0 crossed with SV high, at −0.1130. That is a winner reversal across consistency at fixed surface variability, which the grid pooled over all seven datasets shows only at $\delta = 0.02$. The marginal SV-high band clears the margin by 0.0001, which another bootstrap seed could undo, so we do not count it. With kisti also set aside, every ADS band below the top one and every SV band is a tie, and the top ADS band is a *k-NN win* made entirely of the rule's abstentions.

The reading that the choice barely matters holds for the pool and for inspire, which has four fifths of the blocks, and fails for kisti and zbmath. Where the choice does matter, the two datasets show the paper's two forces at dataset scale: matching, which favours the rule, and coverage, which favours the vote. And under the pre-registered generalization rule, a pattern present in one of four qualifying non-inspire datasets is reported as that dataset's finding and nothing wider, which applies to each of these two. We do not know what makes kisti different. Its agreement rate, 0.7826, is the lowest of any qualifying dataset, so the vote overrides the matched form's own evidence there more often than anywhere else, and it is usually wrong when it does.

7.9 Why no selector experiment is reported

One negative control follows from the numbers above without being run. A design-time selector reading ADS and SV can only route at the granularity those two signals define, which is a cell of the frozen grid, and at the primary margin no cell carries a decided pooled verdict: eighteen of twenty are demonstrated ties and two are undecided. In the pooled grid there is no cell in which one mechanism beats the other by an amount a practitioner would act on, so a selector routing between them would be fitting inside demonstrated ties. We built none. In kisti and in zbmath a selector would have nothing to route either, for the opposite reason: in each, every decided band with thirty or more blocks points the same way, so the best policy inside each is a fixed one.

A selector reading block size instead would route on the one signal whose bands carry a decided verdict in the seven-dataset pool. At fixed consistency the direction never changes, so where consistency is high such a selector reproduces the choice consistency already implies, and where it differs from a fixed choice it trades the rule's precision where it fires for the vote's coverage. That is a policy on abstention, which a practitioner can set directly from the two mechanisms' definitions without measuring block size at all. All of this is a statement about the pre-registered margin and not a claim that the underlying gaps are zero; Section 8.4 sets out what would have to be true for the question to be live.

8 Discussion

8.1 What does not work, and why

Design-time mechanism selection fails here for two reasons that have nothing to do with each other, and separating them is the paper's central contribution. The first is structural: the mechanism pair a practitioner reaches for first is nested, so the winner is not a free variable and there is nothing for a signal to predict. That is a property of the comparison rather than of the corpus or the sample size, and it would hold on any data. The second is empirical and appears only once the first is fixed. With a two-sided pair the mechanisms are close, 0.8244 against 0.8294 pooled, with 10,172 of 13,902 blocks demonstrated ties at a five-point margin and 500 decided either way, and the gap moves with both signals, changes sign on both and stays small on both.

Stated flat, that no design-time signal here is actionable, the claim is too strong, and Section 7.7 is why. Block size decides bands on small blocks at the primary margin, but supplies no reason to prefer one mechanism's matching over the other's: the verdict comes from the vote's full coverage against the rule's abstention, and at fixed consistency block size never points to a different winner. The precise claim is that, pooled over the seven datasets, no design-time signal we examined tells a designer which mechanism resolves a name better.

Neither reason is a measurement failure and both are actionable. The first converts into a check to run before the comparison. The second converts into a cheaper check than any signal: hold out a slice of the same labelled history the signals are computed from, and compare the two mechanisms on it once. On most of our data the difference is inside the margin, and cost and maintainability can decide. On two datasets it is not, and no property of the history we measured says so in advance as plainly as that comparison does. One caveat belongs here and not in the limitations, because it changes the reading: all of this is stated against a five-point margin, and at $\delta = 0.02$ the consistency axis does produce a decided marginal reversal, the rule ahead where the history is most impure and the vote ahead where it names one person. The finding is that no signal is actionable at the margin we pre-registered once the seven datasets are pooled, not that the underlying differences are zero.

8.2 What the structural result means for benchmark and portfolio design

Per-instance algorithm selection assumes a portfolio whose members can beat each other [Rice, 1976, Smith-Miles, 2009, Barbudo et al., 2023]. That assumption is usually left implicit and it is not free: where the members draw their candidate answers from the same history and one is a restriction of another, part of the selection space is empty before the selector is trained, and a selector fitted there is fitting noise. Three cheap practices follow. Check that the members can beat each other per stratum, not only in aggregate; one count does it. Report the structurally empty regions in the design section, where they are a property of the comparison, and not in the results, where they read as a defect. And when the purpose is a comparison, do not calibrate the containing member for accuracy: maximizing its own accuracy walks its parameter toward a limit in which the pair is nested, and the protocol can then certify a nested comparison as a well-calibrated one. The warning applies directly to author name disambiguation, where string-matching baselines are treated as mandatory [Backes, 2018, Kim and Kim, 2020], and to any comparison of rules against retrieval elsewhere.

8.3 What this says about our own earlier result

This paper grew out of an earlier controlled study of ours, a pre-registered 240-condition synthetic factorial comparing exact-match rules against fuzzy retrieval across six consistency targets and two lexical conditions [Sapariya, 2026], a Zenodo-archived research artifact rather than a peer-reviewed publication, which we now regard as superseded by this paper. It reported that realized consistency predicted each mechanism's own accuracy well, Pearson above 0.9 in both lexical conditions, but did not predict which mechanism won, because the winner was instead a constant function of the separately manipulated lexical condition. A frozen consistency-threshold rule agreed with the empirical winner in 100% of comparisons in one consistency band and in 0% in another.

Read through Section 4, part of that shape was not a discovery. The retrieval arm took the single best lexical match over the same historical records the rules arm matched against, so a string present in the history sat at the maximum of the similarity score and retrieval returned what the rules arm returned. The rules arm could differ in its own favour only through tie-breaking among equal-scoring records, or where the two-stage company-then-global fallback both arms shared, an invoice-domain lookup that tries the issuing company's own history before the whole corpus, let them stop at different stages. The containment there is close rather than exact, and close enough: retrieval won every defined comparison and the rules arm won none, while the clean-lexical condition contributed zero defined comparisons. That study's winner rule had no separate label for an undecided comparison. Read against ours, 115 of its 120 clean conditions have intervals inside its two-point margin and the other five cross it on the rules side.

Most of the earlier result stands, and being precise about the part that does not matters. **An artifact of the pair: the constancy of the winner.** "The winner is a constant function of the lexical condition" is what a near-nested pair returns on any data. One value of that constant is retrieval, which the containment makes close to inevitable; the other is, in most conditions, a tie, which that paper's own numbers already say, at a gap of about half an accuracy point in the clean condition against a two-point margin. A consistency signal then looks blind to a choice that was never open, and the 100% against 0% band split follows from a threshold rule agreeing with a constant winner in one band and disagreeing in another. **Not an artifact: the accuracy prediction.** That consistency predicts each mechanism's own accuracy is a claim about each arm separately, and containment says nothing about it; we reproduce it here on real data as H1. **Intact but narrower: the widening margin.** That study reported the rules arm's disadvantage

widening rather than shrinking as consistency rose and presented it as an empirical outcome rather than a consequence of the construction, which was the right call, and we see the same direction on real data. What does not survive is reading a widening margin as evidence about a winner, because in a nested pair the margin can widen without the winner having been in contest at any point. Nothing here re-runs that experiment; its value is that a nested pair produced a clean, publishable-looking shape, and the design rather than the data produced half of it.

8.4 What would have to be true for design-time selection to work

These are conditions, not promises of future work, and any one of them failing is enough to reproduce the result reported here. **The mechanisms have to be far enough apart** that the gap clears a margin the practitioner would act on, and half an accuracy point pooled is not that, whatever its statistical status. **The contested region has to be more than a minority**: here a contest requires a history that is both impure and surface-varied, which is 42.2% of held-out signatures on the evaluation split, and only 500 of 13,902 blocks carry a decided verdict at the primary margin. **The signal's association with the winner has to survive its controls.** Ours had two pre-registered controls, an initialization-free residual and a different metric family, and the surface effect survived both at reduced magnitude. Surviving them is necessary and not sufficient, because an association with the winner is not the same as a gap that clears a margin.

9 Limitations

Power in the contested region, and the margin. Five hundred of 13,902 blocks carry a decided per-block verdict at the primary margin. The margin is what carried both selection hypotheses to Inconclusive, and we do not read Inconclusive as evidence of absence. We kept $\delta = 0.05$ because moving it after seeing that $\delta = 0.02$ is where the structure lives is the move a pre-registration exists to prevent, and a reader holding a different margin can read the whole robustness curve in Tables 4 and 6.

The hypotheses were amended after results existed. They were, and a reader is entitled to hold it against the paper; Section 6.4 states the case. We add that the framing changed twice, once when the hypotheses were amended and once when the amended framing was retired in favour of the result reported here. The second change moves toward the frozen table's own output rather than away from it, and it amends no pre-registration, but it is a change and it is on the record.

One dataset dominates. inspire contributes 11,110 of the 13,902 blocks. Per-dataset and pooled non-inspire reporting was mandatory and was run, and H1 generalizes under the frozen rule. The consistency hypothesis's requirement that its gradient hold in the pooled non-inspire analysis is not met, and no dataset shows a surface-driven winner reversal. Section 7.8 summarizes the band tables in all three forms the pre-registration requires, and `reports/phase2_9b_report.md` gives them in full.

Two simple untrained mechanisms. Section 3.1 gives the reasons for this scope. They are reasons and not excuses, and the scope still bounds the claim: a winner change between a rule and a vote is a narrower phenomenon than one between a heuristic and a learned system, and Kim and Kim tested the harder version and reported no winner change either.

One signal was examined after the fact. Block size was not pre-registered, and Section 7.7 is a supplementary analysis of records that already existed. Its bands are octaves chosen for balance rather than fitted to an outcome and nothing in it changes a hypothesis verdict, but it does not carry the protection the two pre-registered signals carry. The same applies to the sensitivity analysis on blocks with three or more test signatures and to the bootstrap interval on the driver comparison.

The surface axis overlaps with forename initialization. SV correlates with the initialization share at Spearman 0.36. The initialization-free residual arm bounds the overlap rather than removing it: the same shape appears in the residual at smaller magnitude and with no reversal, so the surface effect is more than initialization and is small in both forms.

10 Conclusion

A design-time consistency statistic predicts how hard each mechanism will find a block, with Spearman 0.51 and 0.83 on held-out data across seven datasets. It does not tell you which mechanism to build. Prediction is not selection.

The structural half of that generalizes further than the empirical half. Exact match is a strict sub-case of single-nearest retrieval under any similarity that reserves its maximum for identical strings, so the rule cannot win the comparison between them at any cutoff. More generally a precision-first rule cannot win a block whose history names one author, for any pair of mechanisms that assigns from the block's own history, nor one whose history is written one way, for mechanisms that read surface forms only. Those regions are settled before any data is collected. The empirical half is narrower. Once the datasets are pooled, neither pre-registered signal yields an actionable selection rule at the pre-registered margin. Two datasets depart from the pool, one in each direction. The one clause comparing the two signals found surface stability the stronger driver of decidability, decisively at $k = 5$ and weakly at $k = 2$. Block size, which we did not register, does decide bands at that margin, through the rule's abstention rather than any difference in matching quality, and at fixed consistency it never points to a different winner. Pooled, no design-time signal we examined tells a designer which mechanism resolves a name better.

For a practitioner the reading is short. Check that the two mechanisms you are comparing are not nested, with one count of the items where each is right and the other wrong. Then compare the two once on a held-out slice of your own labelled history before reading any signal: on most of our data the choice did not matter, and on two datasets it did, in opposite directions. If one of the two abstains, settle separately what an abstention costs you, because on small blocks that is what the comparison turns on. And measure your data to learn how hard it is, not to learn what to build.

A Winner structure at all three margins

Reporting a margin where it supports a reading and a different one where it does not is a failure mode a null result is judged on, so the whole surface is here. Table 8 gives the pre-registered outcome label for each stratum of each grid at each of the three margins, where each label summarizes a sequence of the band verdicts defined in Section 3.3: *reversal* when two bands are decided in opposite directions, *tie-crossing* when at least one band is decided, another is a demonstrated tie and none is decided the other way, and *compression* otherwise, which includes a sequence whose decided bands all point the same way with no tie beside them, as at ADS = 1.0 and $\delta = 0.02$. The two reversals are the ones Section 7.5 reports and neither contradicts the statement there that no SV-driven winner reversal exists below ADS = 1.0 in the direction-by-SV grid, which is a different grid.

Table 8: Pre-registered winner-structure outcome by stratum and margin, for the three grids the pre-registration defines. The two reversals are the only ones in these three pooled grids, and both need $\delta = 0.02$. Table 4 holds a third on the marginal ADS bands at the same margin, and Section 7.8 reports decided cells at the primary margin once inspire is set aside. Source: `reports/phase2e_boundary.md`.

Stratum	$\delta = 0.05$	$\delta = 0.03$	$\delta = 0.02$
Across SV bands, at fixed ADS (H3, direction by SV)			
$[0, 0.5)$	compression	tie-crossing	tie-crossing
$[0.5, 0.7)$	compression	compression	tie-crossing
$[0.7, 0.9)$	compression	compression	compression
$[0.9, 1.0)$	compression	compression	compression
ADS $= 1.0$	compression	tie-crossing	compression
Across ADS bands, at fixed SV (H3, non-direction by ADS)			
SV $= 0$	compression	tie-crossing	tie-crossing
SV low	compression	compression	tie-crossing
SV mid	compression	compression	compression
SV high	compression	tie-crossing	**reversal**
Across SV bands, inside each INIT band (secondary control)			
INIT $= 0$	tie-crossing	tie-crossing	**reversal**
$0 <$ INIT < 0.5	compression	compression	compression
INIT ≥ 0.5	tie-crossing	tie-crossing	tie-crossing

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Competing interests

The author declares no competing interests. One disclosure is material to how this paper should be read: Section 8.3 reinterprets an earlier archived research artifact of the author's own, and states there what that artifact is.

Author contributions

Maitreya Sapariya is the sole author and is responsible for the study design, the pre-registration and its amendments, the implementation of the pipeline, the analysis, the interpretation and the manuscript. The role of AI assistance in that work, and its limits, are stated in the generative-AI disclosure below.

Data and code availability

The source code, the pre-registration and its amendments, and the scripts that produce every table and figure in this paper are public in the repository `https://github.com/stiFFLer-codes/`

`prediction-is-not-selection` at tag `v2.0.0`, whose commit history is what Section 6.4 refers to. This preprint is archived on Zenodo at doi:10.5281/zenodo.24184394.

The data is S2AND, the Semantic Scholar Author Name Disambiguation benchmark [Subramanian et al., 2021]. It is publicly available and we use it under its ODC-BY licence. The repository does not carry the raw release. It does carry tables derived from it, redistributed under the same licence with attribution: one row per signature with the name fields, block key, author label and affiliation string, and per-block tables keyed by normalized name. S2AND notes that some affiliation strings come from the Microsoft Academic Graph [Sinha et al., 2015]. The repository also holds the code that downloads the raw release and a manifest with a checksum for every file. Besides the S2AND tables, the repository carries one derived extract of the WDC product-matching corpus [Primpeli et al., 2019], used only for the substrate check in Section 1.2; its source states no licence, and the extract will be removed on request. `phase2/DATA_LICENSES.md` lists every derived file and its terms.

The pipeline is offline and CPU-only. After the one-time S2AND download it makes no network call and needs no GPU and no paid service, and the whole run order, caches included, takes about twenty minutes on the consumer laptop that produced the reported run. Every result here comes from that one machine, at the seed fixed in Section 6.1.

Generative-AI disclosure

The author used an AI assistant (Claude, Anthropic), including the Claude Code agent, throughout: to help plan the study, to draft the pre-registration and its amendments, to write and run the analysis code, to draft and revise this manuscript, and to run a simulated peer review and several audits of it before release, working under author direction. The repository's pass log records that involvement pass by pass, and most of its commits also carry a co-author trailer. The research design, the pre-registration and its amendments, the analyses, every reported number, the selection and verification of every citation, and the final wording were determined, checked and approved by the author, who takes full responsibility for the content. This is a statement about how the paper was produced and not about the systems it studies: both mechanisms evaluated here, the exact-evidence rule and the k-nearest vote, are untrained lookups that invoke no language model.

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